

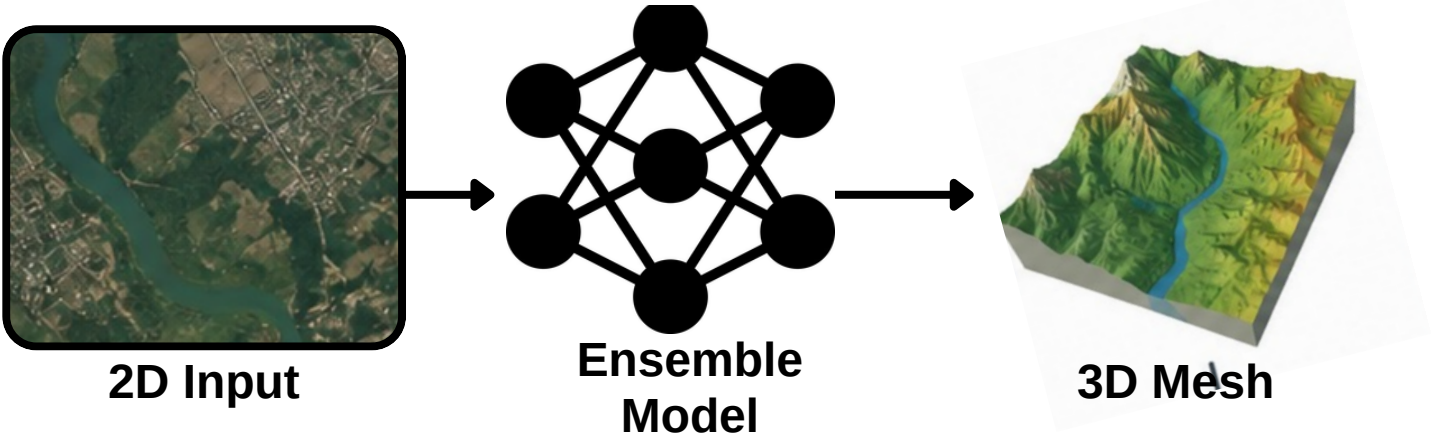
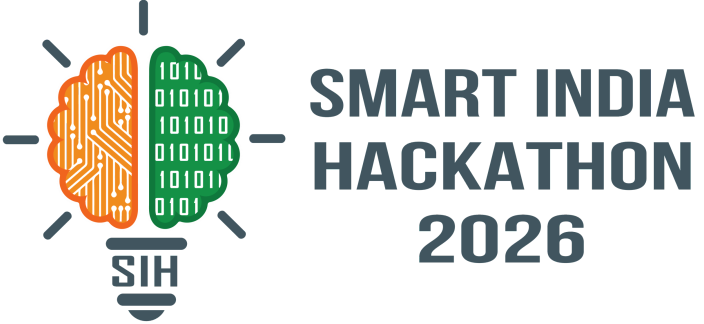
TITLE PAGE

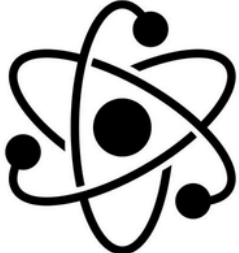
- **Problem Statement ID** – 26175
- **Problem Statement Title** –
DepthWizard Single-View Height
Estimation and 3D Fly-through
- **Theme** – Disaster Management
- **PS Category** – Software
- **Team ID** – Ex101
- **Team Name** – Team Endra





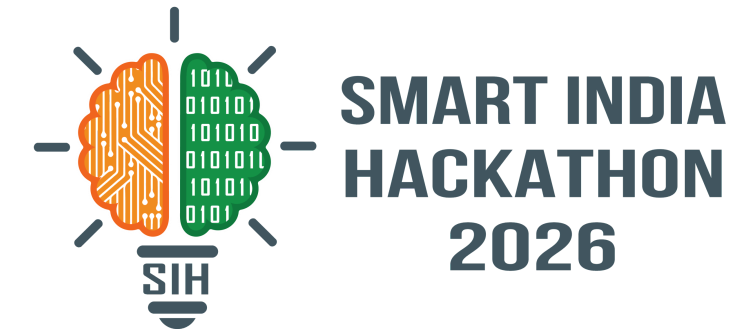
IDEA TITLE



			Supervised Learned Fusion  Dynamic Fusion Weights (Ensemble)	DSM & Flight Simulator  Simulation
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IDEA TITLE



Proposed Solution:

An ensemble of CNN, Transformer, and geometry-aware models for accurate, robust depth estimation.

- Relative & Absolute DSM Generation
- Interactive 3D Terrain Flythrough
- Support for Georeferenced & Non-Georeferenced Images

Approach:

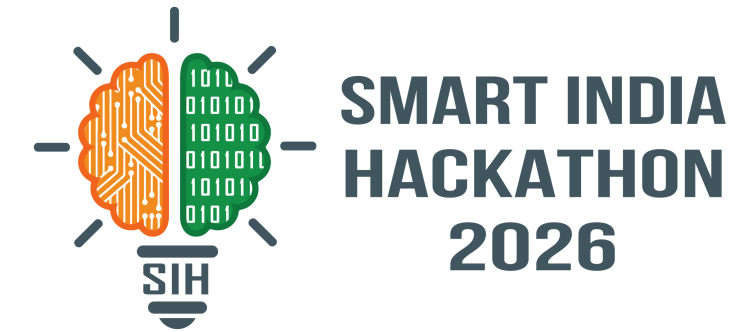
- Learned Fusion of Model Output
- Use of existing geosimulation apps to display results

Novelty:

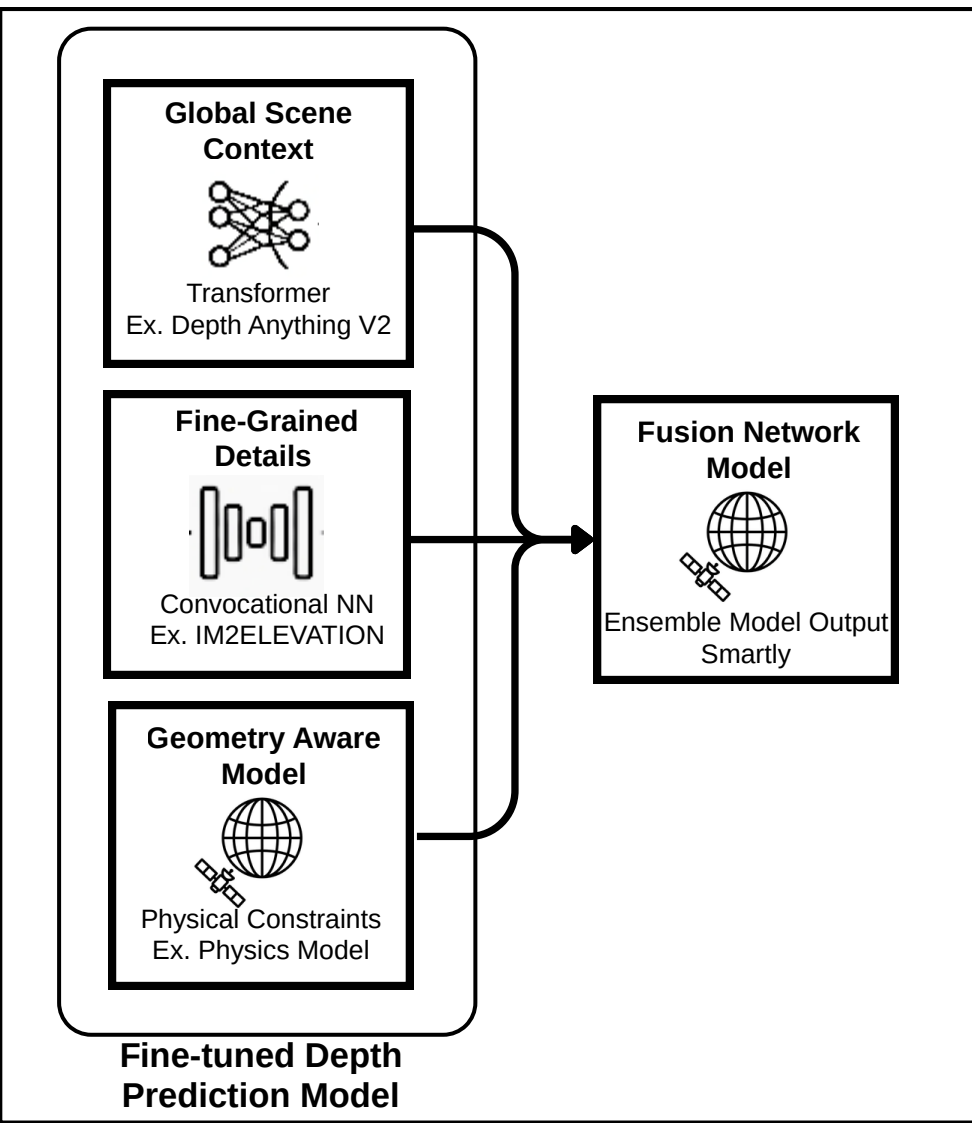
- Monocular Depth Estimation from a single RGB image
- Relative & Absolute DSM generation
- DEM/GCP-based Scale Calibration
- Interactive 3D Terrain Flythrough



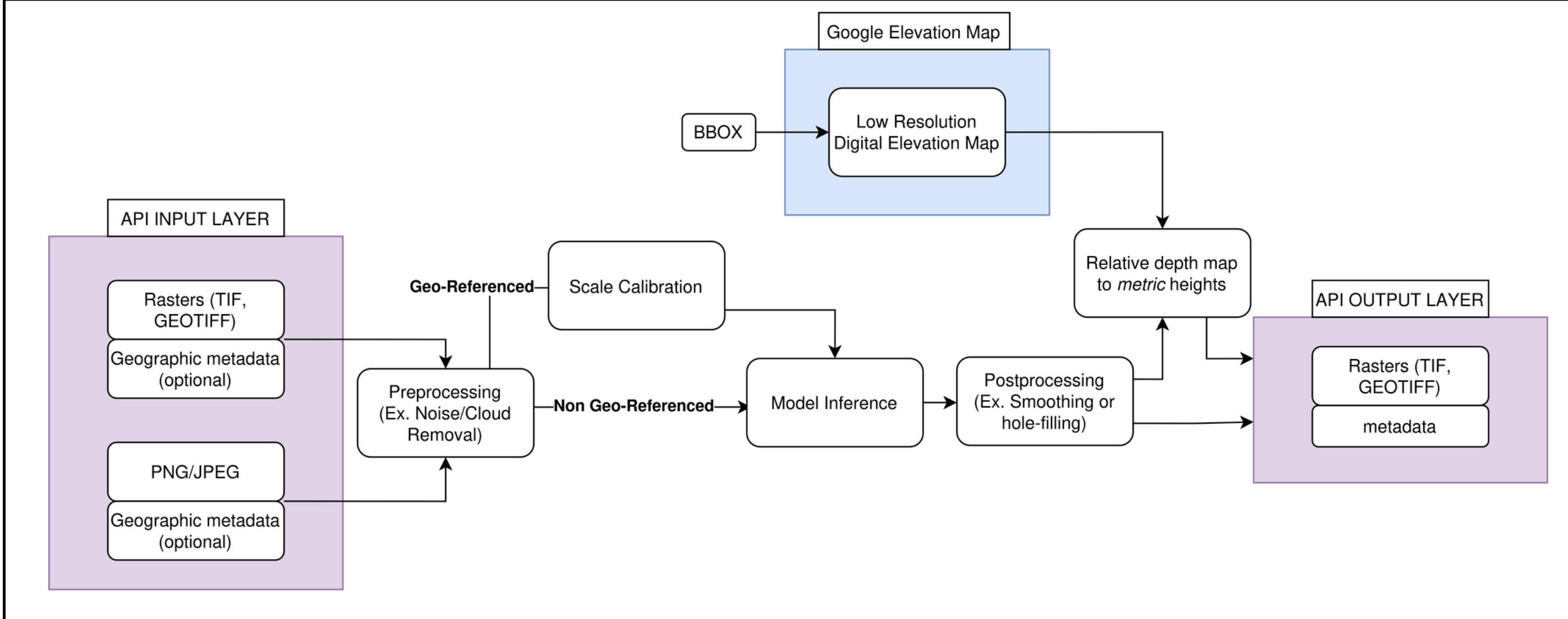
IDEA TITLE



- **Proposed Solution (Describe your Idea/Solution/Prototype)**
- Detailed explanation of the proposed solution
- How it addresses the problem
- Innovation and uniqueness of the solution



Model Inference



Proposed System Architecture

Inputs: the relative depth map, image georeferencing (projective coordinates, possibly camera altitude if available), and ancillary geodata (e.g. low-res DEM or Ground Control Points).

IM2ELEVATION

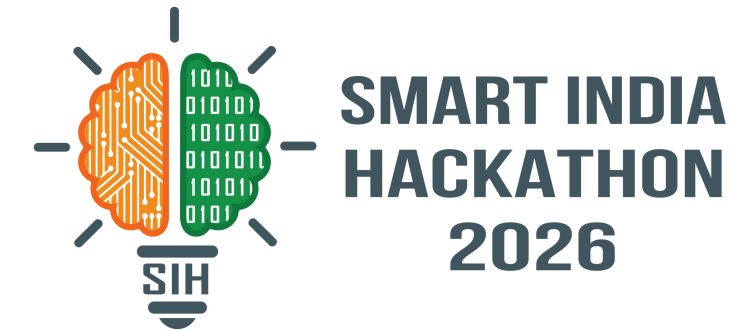
Aerial Image

- Detect building footprint + building shadow**
- Find the shadow length**
- Use the sun elevation angle**
- Calculate building height geometrically**

Evaluation Metrics: Height RMSE and MAE (global and per-scene), and qualitative 3D fidelity (projection error of texture onto mesh). Stability (variance of error) across environments. For visualization: measure of texture-to-geometry registration and user experience.



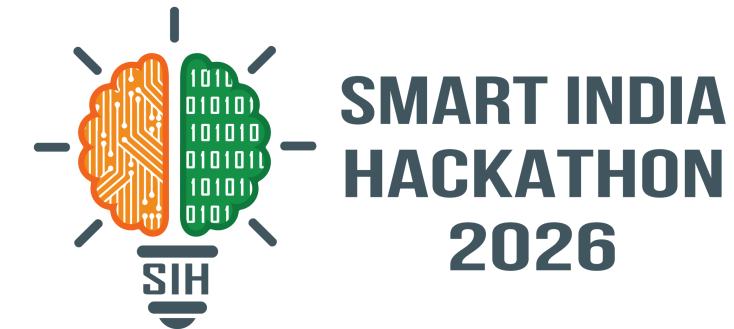
TECHNICAL APPROACH



- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)



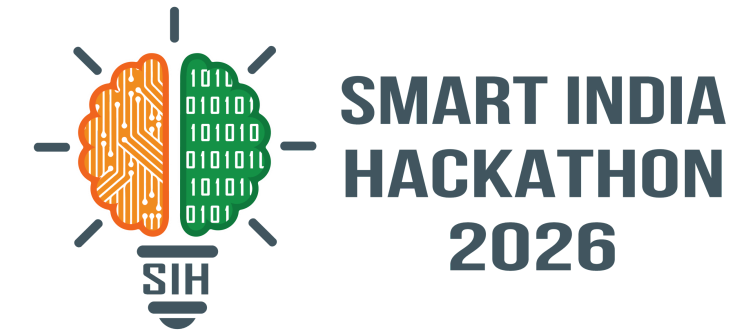
FEASIBILITY AND VIABILITY



Challenges	Solutions
Relative Depth – Scale ambiguous for non-georeferenced inputs	off-the-shelf monocular depth estimator Ex. MiDaS/DPT
Significant Domain Gap – When applied to overhead imagery	off-the-shelf monocular depth estimator Ex. MiDaS/DPT
Spatial Resolution: High required (satellite ~0.3–5 m GSD)	off-the-shelf monocular depth estimator Ex. MiDaS/DPT
Obscuration: Possible clouds, vegetation intervention	off-the-shelf monocular depth estimator Ex. MiDaS/DPT
relative depth: scale-ambiguous results	off-the-shelf monocular depth estimator Ex. MiDaS/DPT



FEASIBILITY AND VIABILITY



- Analysis of the feasibility of the idea
- Potential challenges and risks
- Strategies for overcoming these challenges

Potential Impact:

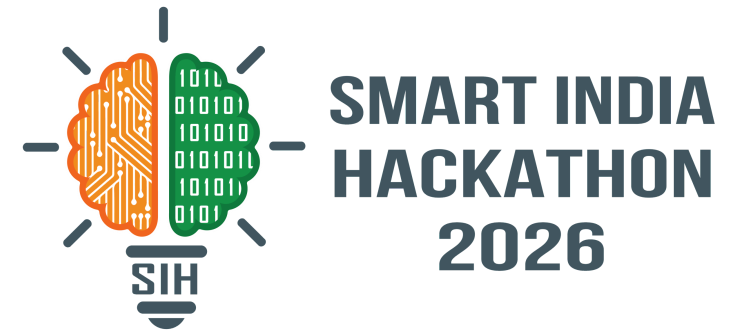
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Benefits:

- **Lower Cost & Rapid Analysis:** XX% savings while XX% Faster Results
- **Hardware-Light:**



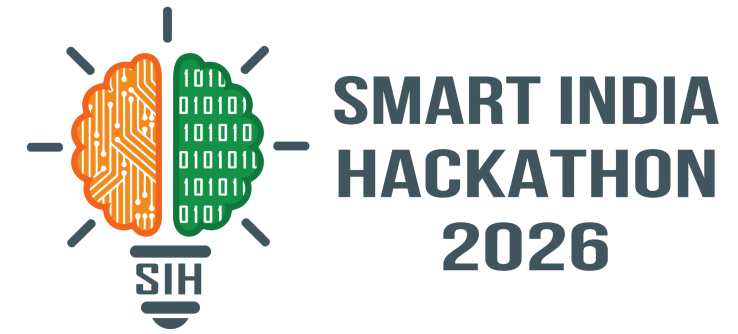
IMPACT AND BENEFITS



- Potential impact on the target audience
- Benefits of the solution (social, economic, environmental, etc.)

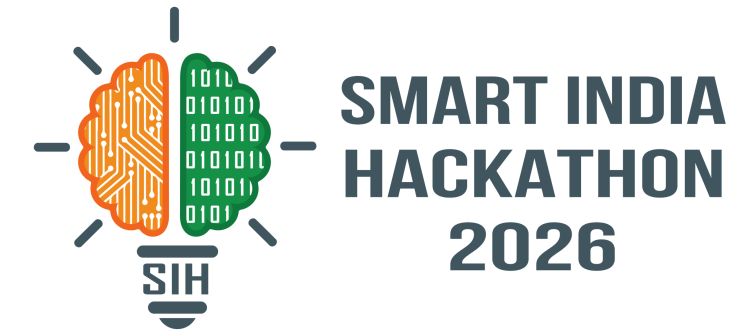


RESEARCH AND REFERENCES



- Details / Links of the reference and research work

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.